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Mid-term Progress Report

Phase-Field Modelling of Crawling Cells

Towards a Tractable Two-Dimensional Model

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Abstract: We implement a continuum phase-field model of eukaryotic cell crawling, coupling a cell field ϕ to an actin polarization $\mathbf{P}$, following Tjhung *et al.* [4] and the effective-adhesion picture of Leoni and Sens [5]. The model is built in four stages so that each physical ingredient can be isolated. A baseline with vanishing fluid velocity translates a cell but cannot deform it and loses 4.5% of its area. Confining treadmilling to a wall layer produces a leading protrusion and a transition in the treadmilling rate. The complete plane model, with Stokes flow, active stress and anchoring, recovers the non-motile actin aster and the lamellipodial fan. Isolating ingredients shows that in this geometry the shape is set almost entirely by anchoring-induced splay of $\mathbf{P}$; the flow contributes about 2%. We therefore replace the elliptic Stokes solve by its friction-dominated limit. The reduced model crawls steadily (travel 239.9 in time 299 at speed 0.801), is about five times faster, agrees with the full calculation to 0.2% where they coincide, conserves volume rather than projected area, and is local — the property needed for many interacting cells.

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Signature of the adviser:

1 Introduction and Motivation

Cell crawling on a substrate underlies wound healing, immune response and metastasis. Mechanically it combines actin treadmilling at the front, myosin-driven contraction at the rear, and focal adhesions that couple the network to the substrate [1]. Keratocyte fragments that lack the nucleus still polarise and crawl [2], and motile shapes can be predicted from a few mechanical parameters [3]. This motivates minimal physical models.

Tjhung, Tiribocchi, Marenduzzo and Cates [4] treat a crawling cell as a droplet of active polar fluid and, in three dimensions, obtain a fried-egg aster, a keratocyte-like lamellipodium and a finger-like pseudopod. Averaged over stochastic mechanosensitive linkers, substrate adhesion reduces to an effective friction [5], which is how it enters depth-averaged continuum models [6]. The long-term goal of this project is many interacting cells. The Stokes solve of Ref. [4] is elliptic, hence expensive for one cell and prohibitive for many. The mid-term work therefore implements the full model, identifies which ingredients actually matter, and constructs a cheaper model whose accuracy we can quantify (Table 1).

Ver.	Geometry	Force balance	Area
1	plane, free space	$\mathbf{v} = 0$	drifts 4.5%
2	slice with a wall	screened in z	fixed exactly
3	plane	incompressible Stokes	fixed exactly
4	plane	friction dominated	free, soft

Table 1: Four stages of the model. Versions 2 and 3 are complementary (side view vs. top view). Version 4 is the tractable model for later multi-cell work.

2 Computational Setup

The cell is a diffuse field ϕ (1 inside, 0 outside); $\mathbf{P}$ is the coarse-grained actin polarization; $\mathbf{v}$ is the cytoplasmic velocity. Following Ref. [4],

$$F = \int d^2r \left\{ a\phi^2(\phi - 1)^2 + \frac{\varepsilon^2}{2} |\nabla\phi|^2 - \frac{\alpha}{2} (2\phi - 1) |\mathbf{P}|^2 + \frac{\alpha}{4} |\mathbf{P}|^4 + \frac{\kappa}{2} (\nabla\mathbf{P})^2 + \beta \mathbf{P} \cdot \nabla\phi \right\}. \quad (1)$$

The first two terms generate interfacial tension. The α terms drive $|\mathbf{P}|^2 = 2\phi - 1$ inside the cell only. The Frank term aligns neighbouring actin. Anchoring $\beta > 0$ points $\mathbf{P}$ outward, normal to the boundary, and is the controlling shape parameter. Transport is by $\mathbf{u} = \mathbf{v} + w_0\mathbf{P}$, with w_0 the treadmilling (polymerisation) speed:

$$\partial_t\phi + \nabla \cdot (\phi\mathbf{u}) = -M[\mu_\phi - \Lambda g(\phi)], \quad \partial_t\mathbf{P} + (\mathbf{u} \cdot \nabla)\mathbf{P} = -\boldsymbol{\Omega} \cdot \mathbf{P} + \chi\mathbf{D} \cdot \mathbf{P} - \mathbf{h}/\Gamma, \quad (2)$$

where $\mu_\phi = \delta F/\delta\phi$, $\mathbf{h} = \delta F/\delta\mathbf{P}$, $g = \phi(1 - \phi)$, and $\mathbf{D}$, $\boldsymbol{\Omega}$ are the strain-rate and vorticity. At Reynolds number $\sim 10^{-5}$ the force balance is Stokes. In a top view the wall of Ref. [4] becomes a depth-averaged friction $-\xi\mathbf{v}$ [5],

$$0 = -\nabla p + \eta\nabla^2\mathbf{v} - \xi\mathbf{v} + \mathbf{f}, \quad \nabla \cdot \mathbf{v} = 0, \quad (3)$$

with $\mathbf{f} = \nabla \cdot (\sigma^{\text{act}} + \sigma^{\text{el}}) - \phi\nabla\mu_\phi$ and $\sigma_{\alpha\beta}^{\text{act}} = -\zeta\phi P_\alpha P_\beta$ ($\zeta < 0$ contractile). Version 3 solves (3) spectrally. Version 4 uses $\eta/(\xi R^2) \approx 0.006$ for $R \approx 13$: friction, not viscosity, balances the active forces, leaving the local Darcy law $\mathbf{v} = \mathbf{f}/\xi$.

Versions 2 and 3 enforce rigid area with a spatially uniform Lagrange multiplier $\Lambda = \int \mu_\phi / \int g$ [7]. A real cell has roughly constant *volume* $V_0 \simeq hA$, so spreading is allowed if paid for by thinning; Version 4 uses the soft law $\Lambda = -k_A(A/A_0 - 1)$. Advection is written in conservative flux form on a periodic grid and advanced by explicit Euler.

3 Results and discussion

Version 1. With $\mathbf{v} = \zeta = \beta = 0$ a disc translates but cannot change shape (uniform $\mathbf{P}$ gives uniform $\mathbf{u}$) and loses 4.5% of its area to non-conservative advection.

Version 2: protrusion at the wall. The thin protrusion of Ref. [4] is a near-substrate structure, so we work in a vertical (y, z) slice with $w(z) = \frac{w_0}{2} [1 - \tanh((z - \lambda)/w_\tau)]$ and the Brinkman closure $(1 - \ell_v^2 \partial_z^2)\mathbf{u} = w(z)\mathbf{P}$, $\ell_v = \sqrt{\eta/\xi}$. Near-wall material runs out ahead of the body; tension pulls it back; that competition is the protrusion (Fig. 1a,b). At $w_0 = 0.3$ the cell stays a hemispherical cap (contact length 44.0, speed 0.050); at $w_0 = 2.0$ it draws a thin leading layer (contact 63.0, speed 0.432). Volume is conserved to 10^{-16} .

Sweeping w_0 (Fig. 1c) shows the protrusion transition: the contact length is nearly flat below $w_0 \approx 0.8$ and then rises steeply, close to $M\varepsilon^2/\lambda \approx 0.75$. The lamella length and crawl speed rise in the same window.

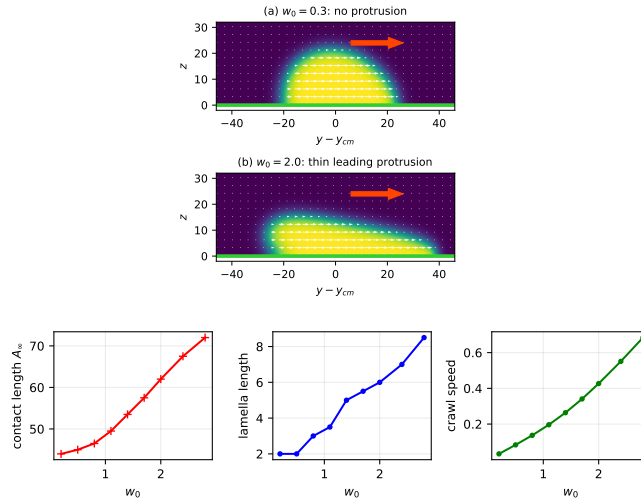


Figure 1: Version 2. Top: vertical slice, wall at $z = 0$ (green); (a) $w_0 = 0.3$, hemispherical cap; (b) $w_0 = 2.0$, thin leading protrusion. Arrows: $\mathbf{P}$. Volume error 10^{-16} . Bottom: contact length, lamella length and crawl speed versus w_0 .

Version 3: shapes in the plane. The catalogued shapes of Ref. [4] are top views, so Version 3 solves the complete plane system. The passive limit is verified first: an ellipse relaxes to a circle (aspect $1.689 \rightarrow 1.010$), the free energy decreases, the flow dies and the volume error is exactly zero.

With activity (Fig. 2, top), strong outward anchoring locks $\mathbf{P}$ into a radial aster: $|\langle \mathbf{P} \rangle| = 0$ and the speed is zero to machine precision, the fried egg of Ref. [4] and of experiment [8]. Moderate anchoring gives a fan elongated *across* the path (aspect 2.91, speed $0.843 \approx w_0$), the keratocyte lamellipodium. Driving anchoring harder gives a more extreme fan (aspect 4.35). A plane pseudopod cannot form (contractility only widens the cell) and the phagocytic cup is three-dimensional; Version 2 already captures the vertical mechanism of the former.

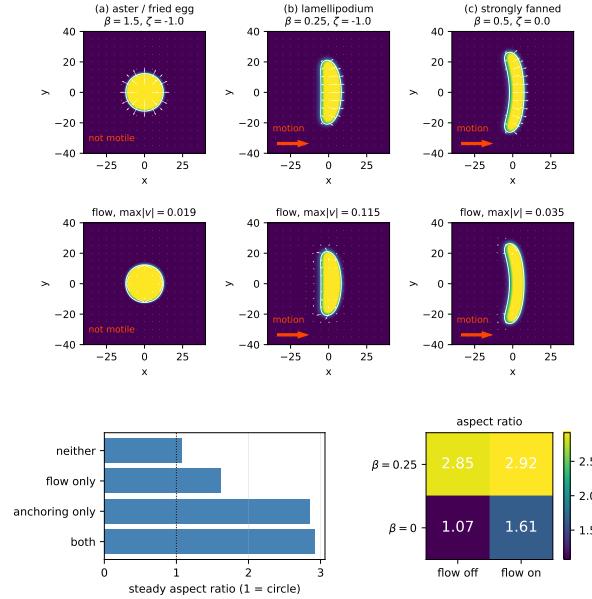


Figure 2: Version 3. Top: morphologies at $w_0 = 1 - \phi$ and $\mathbf{P}$ (upper row) and flow $\mathbf{v}$ (lower row). (a) Aster, not motile. (b) Lamellipodial fan, aspect 2.91, speed 0.843. (c) Stronger fan, aspect 4.35. Bottom: 2×2 of anchoring vs. flow; anchoring alone $1.07 \rightarrow 2.85$, both together 2.92 (+2%).

A 2×2 control (Fig. 2, bottom) isolates the two shape mechanisms. Neither: translating disc, aspect 1.07. Flow alone: 1.61. Anchoring alone: 2.85. Both: 2.92 ($\approx 2\%$ above anchoring). Speed is 0.80–0.85 in every case, set by $w_0\langle\mathbf{P}\rangle$. Sweeping ξ over a 32-fold range screens the flow as $1/\xi$, leaves the speed almost unchanged (0.829–0.845), and saturates the aspect onto the flow-free value 2.85. The flat speed disagrees with the biphasic curve of Ref. [5]: there propulsion is traction and requires mechanosensitive bonds, which a constant ξ cannot see.

Version 4: a crawling cell without Stokes. The Darcy reduction is local and algebraic. Only the active part $\mathbf{v} = -(\zeta/\xi)\nabla \cdot (\phi\mathbf{P}\mathbf{P})$ is kept (vorticity and flow alignment change the shape by $< 1\%$). Contractility is recalibrated to $\zeta = -0.25$ (without pressure the same ζ is $\sim 10\times$ too strong).

Figure 3 is the main result. In the laboratory frame the cell travels 239.9 length units in time 299 at a steady speed 0.801: the circular seed becomes a fan elongated across its path and then crawls in a straight line. Cell-centred snapshots show ϕ and the mean polarization at $t = 0, 100$ and 299. The centre of mass advances linearly in x with $y_{\text{cm}} = 0$; the speed is constant; the aspect ratio saturates near 3.2; A/A_0 grows by $\approx 15\%$ and the implied thickness $h = V_0/A$ falls by the same factor.

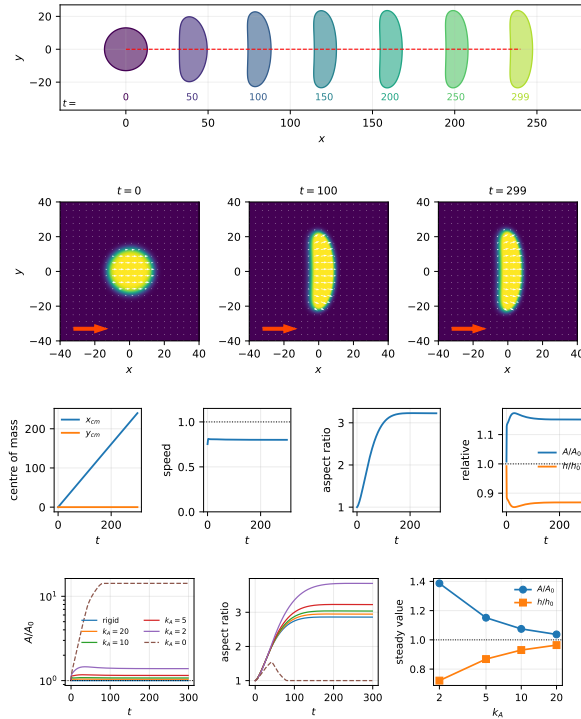


Figure 3: Version 4 crawling (the main result). (a) Laboratory-frame outlines coloured by time; travel 239.9 at speed 0.801. (b) ϕ at $t = 0, 100, 299$ (red arrow: mean $\mathbf{P}$). (c) x_{cm} (linear), speed (0.801), aspect ratio ($\rightarrow 3.2$), A/A_0 and h/h_0 . (d) Soft volume: $A/A_0(t)$ and aspect ratio for several k_A ; $k_A = 0$ inflates $\sim 14\times$.

Where Versions 3 and 4 coincide (flow off, rigid area) they agree to 0.2%. Soft volume ($k_A = 5$) changes speed by $< 1\%$ and shape by 35% (Fig. 3d). Removing the constraint ($k_A = 0$) inflates the cell 14-fold and stops crawling. Cost falls from 2.70 to 1.10 ms/step and the stable dt rises from 0.01 to 0.03, about $5\times$ overall.

4 Conclusions and future scope

The four stages are verified. Volume is conserved to machine precision where it should be. A substrate-confined treadmilling layer produces a protrusion and a transition at $w_0 \approx M\varepsilon^2/\lambda$ (Fig. 1). The plane model recovers the aster and the lamellipodial fan (Fig. 2); shape is anchoring-driven (+2% from flow). Version 4 crawls at speed 0.801 (Fig. 3), is $\approx 5\times$ faster,

agrees to 0.2% where comparable, and is local. Soft volume lets the cell spread and thin; $k_A = 0$ is not viable. Next: two cells, collisions, spontaneous polarity, load-dependent friction [5], then collectives.

Acknowledgements

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